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What Generalizations About English Poetry Survive Measurement? A Distant Reading of Approximately 34,000 Poems, 1600–1925

Samyukta Twarakavi

Abstract

Literary history has often advanced confident, frequently repeated generalizations of the evolution of English poetry: that the Romantics turned to nature, that verse grew more secular around the nineteenth century, that industrialization and the industrial revolution introduced machinery into poems, and that the modern era is darker and sadder than what preceded it. The goal of this study is to test the strength of these claims when operationalized and tested across tens of thousands of poems. This was done using a de-duplicated corpus of approximately 34,000 public-domain English poems (sourced from PoeTree and PoetryDB), each dated by its author's active year and binned into 25-year eras from 1600 onward. Eight named historical claims are translated into pre-poem measures: word-list theme rates for nature, sacred and industrial vocabulary, and lexicon based scores for concreteness, valence, arousal, visual imagery and sadness. Each measure is averaged per era with 95% confidence intervals obtained by bootstrapping whole poets, and each claim is assigned a verdict: The secularism and turn to concreteness claims are confirmed; the Romantic nature turn, industrial imagery, and visual turn are weakly supported (correct direction, overlapping intervals); and two generalizations are challenged, because the measured trend runs opposite to the prediction: in this corpus, modern poetry exhibits neither exhibits higher emotional arousal nor a greater rate of explicitly sad vocabulary than mid-eighteenth century verse. The measures are validated against published lexicons (Brysbaert Concreteness, NRC-VAD) and through convergent-validity checks; lexicon coverage is reported; and the failure cases—including one convergent validity check that resolves in the unexpected direction—are treated as the substantive contribution. All measures are descriptive: no causal claim is made. The corpus is public and the pipeline is fully reproducible.

Keywords

Distant reading; computational literary studies; cultural analytics; text mining; English poetry; word-list theme rates; lexicon-based measures; bootstrap confidence intervals

1. Introduction

Distant reading proposes that certain questions about literary history are better addressed by counting across many texts rather than paying attention to a few (Moretti 2013; Jockers 2013).

Where a traditional literary historical argument advances a generalisation (such as “the Romantics wrote about nature” or “poetry secularized”), distant reading can ask whether that generalisation holds a measurable tendency across a large, dated corpus (Underwood 2019; Piper 2018). The informative outcomes are not only confirmations (which demonstrate that an inexpensive measure can recover a known pattern), but the cases in which the data does not agree, they are a confident claim that does not survive measurement, is itself a literary-historical result.

This study takes eight specific, named claims from the standard narrative of English poetry, operationalizes each as a single number, and tests it against the data. Each claim is expressed as a triple: a measure, an “after” era to which the claim refers, and a predicted direction to an earlier “before” era. The claims are: (1) the Romantic nature turn—nature vocabulary is higher around the 1800s than in the mid 1700s; (2) secularisation—sacred vocabulary falls across the 1800s; (3) the turn to concreteness—poetic language becomes more definite after approximately 1800; (4) industrial imagery—industrial and urban vocabulary rises through the 1800s; (5) reduced valence—a darker, modern mood rises into the twentieth century; (6) rise in arousal—greater emotional intensity into the twentieth century; (7) the visual turn—visual imagery rises, peaking around Imagism (approximately around 1910–1920); and (8) a sadder modern mood—explicit sadness vocabulary rises into the twentieth century.

These claims are not invented for the present exercise; each is drawn from the standard literary-historical record. The Romantic-nature turn is a commonplace of studies on Romanticism (Abrams 1953, 1971; Williams 1973; Bate 1991). The secularisation thesis follows Abrams (1971) and the broader historical accounts of Chadwick (1975) and Taylor (2007). The turn to concreteness is the explicit programme of Imagism (Pound 1913; Hulme 1924) and has already been measured, as an abstract-to-concrete prose across nineteenth century prose, by Heuser and Le-Khac (2012). Industrial imagery is treated by Sussman (1968) and Williams (1973), and the visual turn is associated with the image-centered poetics of Imagism (Pound 1913; Kenner 1971). The remaining claims—a darker, more intense and sadder modern mood—are more diffuse received intuitions than precisely attributable theses; notably, the relevant computational work points the other way, reporting a long decline in emotional, and especially positive-emotional, vocabulary (Acerbi et. al 2013; Morin and Acerbi 2017). The present study counts poetic vocabulary in the tradition of Miles (1957) and of distant reading more broadly (Moretti 2005; Jockers 2013; Underwood 2019).

Every measure is either a word-list rate (the share of a poem’s words belonging to a hand-built theme list) or a mean lexicon score (average of quantitative values generated to measure the emotion, polarity, or subjectivity of a text). The objective is not methodological novelty but a transparent, reproducible audit of received claims, accompanied by honest confidence intervals and an explicit validation of the instruments used to measure.

2. Corpus

The corpus combines two public-domain sources. The first is the English subset of PoeTree (Zenodo record 10907309), a large collection in which each poem carries an embedded author block (name, birth, death) and a body of lines. The second is PoetryDB (poetrydb.org) which contributes further canonical public-domain poems but provides no dates. PoeTree poems are dated by their recorded year of creation where present, and otherwise by the author’s birth year plus a fixed floruit offset of

35 years; PoetryDB poems, lacking any date, are dated by their author’s active year drawn from a hand-curated author-to-year table. Poems shorter than 20 words are removed, as are poems flagged as duplicates within PoeTree and any per-author excess beyond a cap, and the two sources are merged on a normalised fingerprint of each poem’s opening words so that poems appearing in both are counted once. The merged file contains approximately 34,000 poems.

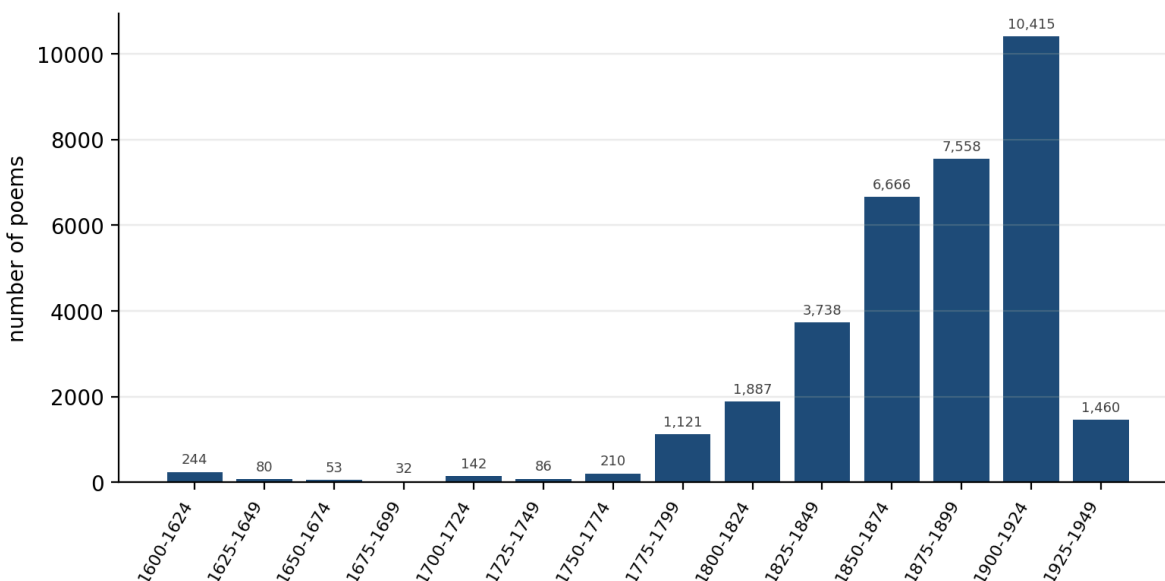


Figure 1. Number of poems per 25-year era in the analysis sample (1600 onward). The corpus was thin before 1775 and concentrated in the nineteenth and early twentieth centuries.

Each poem is assigned to a 25-year era by its date. Analysis begins at 1600: earlier bins are sparse and written in middle English, which distorts any token-based measure. After this cutoff, the analysis sample comprises 33,692 poems across 14 eras (1600–1624 through 1925–1949). The corpus is heavily weighted towards the long nineteenth century and after; per-era counts rose from a few dozen poems in the seventeenth century bins to 10,415 poems in 1900–1924 (Figure 1). This imbalance bears directly on interpretation—early eras carry wider confidence intervals—and is reported explicitly rather than smoothed over. Two properties of the corpus bound what it can demonstrate. It is canonical and public domain, so it terminates in the early twentieth century and over-represents anthologised poets. Along with that, dating is by author floruit rather than composition date, so a long-lived poet’s early and late work share the same date. Both limitations are revisited in section 7 (Limitations).

3. Method

3.1 Measures

The text is lowercased and split on non-letter characters before scoring. Three measures are word-list rates: the percentage of the poem’s tokens appearing in a hand-built list for nature (flora, fauna, weather, landscape, water, seasons), sacred (the divine, the afterlife, worship, sin and holiness, scripture) and industrial (machines and parts, steam and fuel, metal, factory and works, transport and the industrial city). These three themes have no lexicon equivalent, so the lists are kept broad and balanced across sub-themes, including singular and plural forms, and deliberately

exclude words that double as common verbs (ex. Pine, swallow, train) in order to keep the counts conservative.

Five further measures use published lexicons when available, falling back to word lists otherwise: concreteness, scored as the mean Brysbaert concreteness rating of a poem's covered words (Brysbaert, Warriner and Kuperman 2014), roughly 1 (abstract) to 5 (concrete); valence, arousal, and dominance, scored as mean NRC-VAD values (Mohammad 2018); visual and auditory imagery, scored as mean perpetual-strength ratings from the Lancaster Sensorimotor Norms (Lynott et al. 2020); and joy, sadness, and fear, scored as the rate of words tagged with the relevant emotion in the NRC Emotion Lexicon (EmoLex; Mohammad and Turney 2013). Where a measure uses a polarity word list rather than a lexicon, it is scored as the difference between the high and low counts divided by their sum. A poem containing none of a lexicon's words is recorded as missing for that measure and excluded from its era averages rather than counted as zero to avoid any distortions to the results.

3.2 Per-era averages and confidence intervals

For each measure the mean is computed over all scored poems in an era. Confidence intervals are obtained from a bootstrap that resamples whole poets rather than individual poems (500 replicates): within an era, poets are drawn with replacement and all of a drawn poet's poems are included, and the 2.5th and the 97.5th percentiles of the resampled era define the 95% interval (Efron and Tibshirani 1993). Resampling poets rather poems prevents a single prolific poet from dominating an era and yields intervals that reflect between-poet variation, which is the relevant uncertainty for a claim concerning an era as a whole.

3.3 Verdicts

For each claim, the mean of the “after” era is compared to that of the “before” era and a verdict is assigned. A claim is confirmed when the change is in the predicted direction and the two eras 95% confidence intervals do not overlap; weakly supported when the direction is correct but the intervals overlap; and challenged when the change is in the wrong direction. The scheme is deliberately conservative: “weakly supported” covers the common case in which the sign is correct but the corpus cannot exclude no change, and “challenged” is reserved for trends that actively contradict the claim.

4. Results

4.1 Scoreboard

Table 1 reports the verdict for each claim with the before and after era means. Table 2 adds 95% intervals for the claims discussed in the text, and Figure 2 displays them.

Claim	Measure	Before	After	Verdict
Romantic nature turn	nature	1750–74: 2.547	1800–24: 2.980	Weakly supported

Claim	Measure	Before	After	Verdict
Secularisation	sacred	1800–24: 1.443	1900–24: 0.960	Confirmed
Turn to concreteness	concreteness	1750–74: 2.569	1900–24: 2.666	Confirmed
Industrial imagery	industrial	1775–99: 0.147	1875–99: 0.188	Weakly supported
Darker modern mood	valence	1750–74: 0.118	1900–24: 0.108	Weakly supported
Greater emotional intensity	arousal	1750–74: -0.048	1900–24: -0.077	Challenged
Visual turn	visual	1750–74: 2.393	1900–24: 2.436	Weakly supported
Sadder modern mood	sadness	1750–74: 3.054	1900–24: 2.725	Challenged

Table 1. Claim scoreboard. Values are era mean measure scores

Claim	Before era (95% CI)	After era (95% CI)
Secularisation (sacred)	1.443 [1.201, 1.661]	0.960 [0.861, 1.068]
Turn to concreteness	2.569 [2.526, 2.632]	2.666 [2.652, 2.681]
Romantic nature turn	2.547 [1.733, 3.357]	2.980 [2.586, 3.292]
Industrial imagery	0.147 [0.111, 0.182]	0.188 [0.169, 0.207]
Emotional intensity (arousal)	-0.048 [-0.056, -0.043]	-0.077 [-0.082, -0.072]
Sadder modern mood (sadness)	3.054 [2.548, 3.416]	2.725 [2.624, 2.817]

Table 2. Selected claims with 95% bootstrap intervals (by-poet resampling)

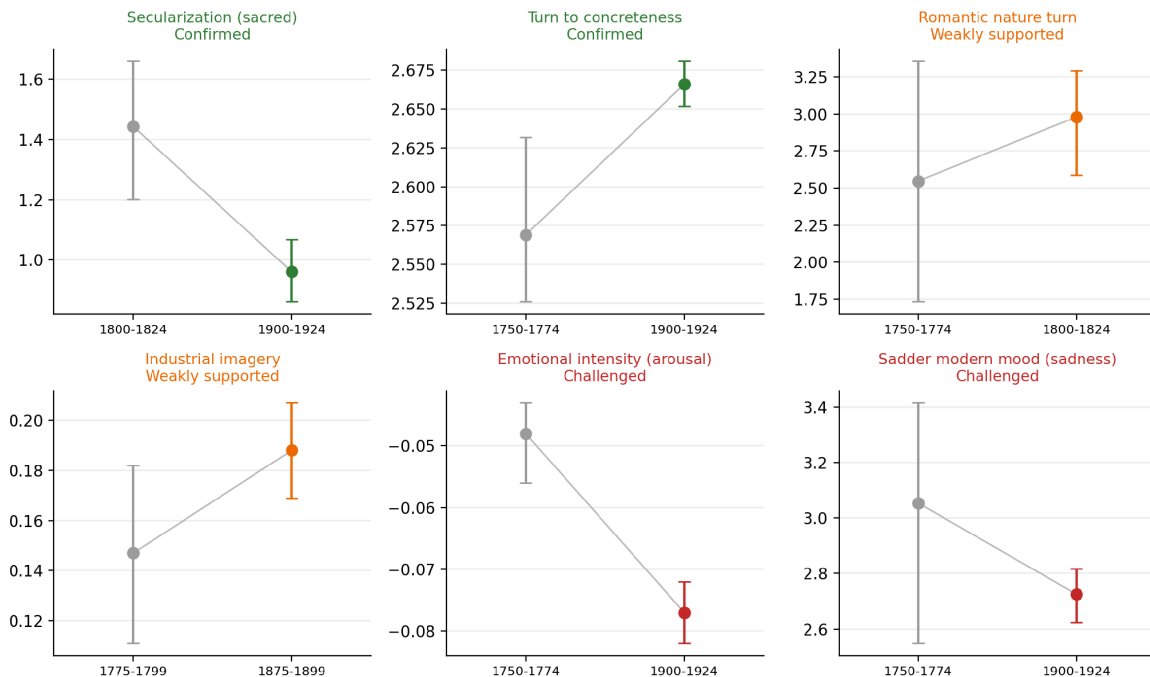


Figure 2. Before (grey) and claim area (colored) means with 95% bootstrap intervals, for the six claims with interval estimates available. Color encodes the verdict: green = confirmed, orange = weakly supported, red = challenged

4.2 The confirmations

Two claims are confirmed with non-overlapping intervals. Secularisation is the clearest: the mean sacred word rate falls from 1.44% in 1800-1824 to 0.96% in 1900-1924, a decline of roughly one third whose intervals are well separated. The turn to concreteness also holds: mean Brysbaert concreteness rises from 2.569 to 2.666 between the mid-eighteenth century and the early twentieth, and because concreteness is averaged over nearly every word in each poem (high lexicon coverage; Section 5), its era estimates are tight and the intervals do not overlap. The absolute change is small and well-estimated.

4.3 The weak points

Four claims point in the predicted direction but cannot clear the bar. The Romantic nature turn rises as expected (2.55 to 2.98), but the “before” era (1750–1774) contains only about 210 poems, therefore its interval is correspondingly wide ([1.73, 3.36]), subsuming the difference. Industrial imagery rose from 0.15% to 0.19% across the nineteenth century—a genuine increase, but on a very small base and with intervals that barely separate. The darker modern mood (valence) drifts downward in the predicted direction but well within overlap, and the visual turn rises only marginally (2.393 to 2.436). In each case the sign agrees with the literary history while the magnitude is too small, or the early era data is too sparse, for the corpus to exclude no change.

4.4 The challenges

Two claims fail outright because the measured trends run against the prediction. Emotional intensity was expected to rise into the modern era; mean arousal instead falls (-0.048 to -0.077). Explicit sadness was expected to rise; the sadness word rate instead declines (3.05% to 2.73%). Both reversal must be read with a serious caveat established in the next section—the emotion lexicon covers only a small fraction of the corpus—but taken at face value they indicate that, in this canonical corpus and on these measures, modern poetry is neither more intense nor more openly sorrowful than mid-eighteenth century verse. This is precisely the kind of result distant reading exists to surface: a confident and intuitive claim that the counting does not support.

4.5 Commented examples

To confirm that the measures track what a reader would observe, the highest- and lowest-scoring poems per-measure were inspected (restricted to poems of at least 40 words, whose rates are more stable). The extremes are legible. The highest-nature poems are dense catalogues of flora and weather—a Laurence Alma-Tadema piece (1900-1924) opening on apples, robins, oats, and sheaves scores roughly 28% nature words—whereas the lowest are martial or doctrinal poems with no landscape at all. The highest-industrial poem is Arthur Stringer’s foundry scene of furnaces, molten metal, and cranes; the lowest are pastoral. The highest-concreteness poems are physical and imagistic, the lowest are concerning thought, soul, and heaven. The highest visual poem (Elinor Wylie) is constructed from explicit contrasts of color and light, while the lowest are sound driven. These extremes confirm that each measure captures the dimension it claims to, rather than an artifact.

5. Validation (failure-case analysis)

As the measures are inexpensive, they require explicit validation. Three checks are reported.

5.1 Word-list rulers versus published lexicons

Concreteness and valence possess both a hand-built word list and a published lexicon, allowing the two independent instruments to be correlated. Concreteness agrees strongly at the level relevant to claims—era-level $r = 0.84$ between the word list and the Brysbaert lexicon (poem-level $r = 0.34$, expectedly lower given per-poem noise). Valence agrees only moderately (era-level $r = 0.41$; poem level $r = 0.44$). The valence instrument is therefore weaker than the two, and the valence-based verdict (the darker, modern mood) should be read accordingly.

5.2 Lexicon coverage

Coverage is the share of corpus tokens a lexicon scores; a low-coverage lexicon yields a noisier measure. Coverage is high for concreteness (Brysbaert, 85.4%), the Lancaster imagery norms (85.4%), and NRC-VAD (68.9%). It is very low for the NRC Emotion Lexicon (7.1% of corpus tokens; Figure 3). This is the single most important caveat in the study: the joy, sadness, and fear measures rest on a thin slice of the vocabulary, so the two challenged emotional claims (Section 4.4) may reflect lexicon sparsity as much as any genuine literary trend. The limitation is reported rather than concealed; a higher coverage emotion lexicon is the obvious next step.

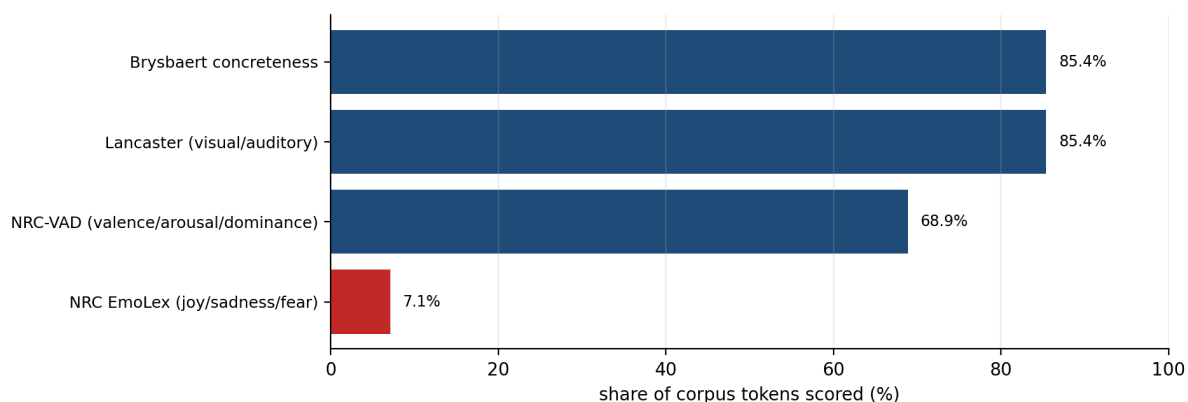


Figure 3. Lexicon coverage as a share of corpus tokens. The NRC Emotion Lexicon (red) covers only 7.1% of tokens, far below the other resources.

5.3 Convergent validity

The lexicon-only measures (arousal, dominance, imagery, emotions) have no second independent instrument, so each is instead checked against trusted measures in the theoretically expected direction (figure 4). Five of six checks pass: sad-word poems score lower valence ($r = -0.40$). Joy-word poems have higher valence (+0.53), fear-word poems lower valence (-0.40) and higher arousal (+0.38), and visual imagery tracks concreteness strongly (+0.73). One check fails: joy and sadness, expected to pull in opposite directions, correlate weakly positive ($r = +0.07$). The most plausible reading, consistent with the 7.1% coverage reported above, is that a poem rich in any emotional vocabulary tends to register on both lists simultaneously, so the EmoLex rates behave

partly as a generic emotionality signal rather than cleanly opposed poles. This failed check is reported as a flag on the emotional measures, not explained away.

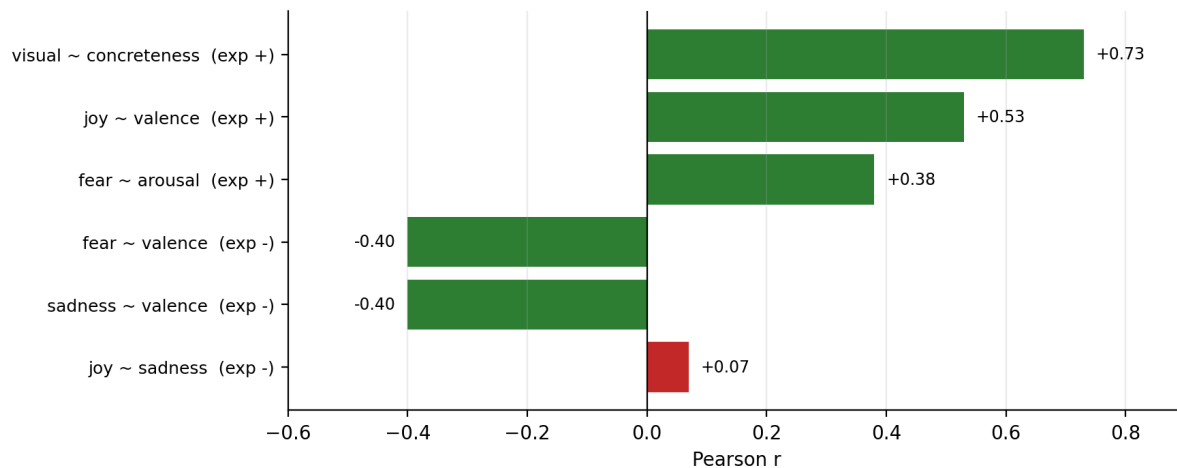


Figure 4. Convergent-validity checks. Bars show the Pearson correlation between each lexicon-only measure and a trusted measure; green indicates the expected direction, red the single unexpected result (joy versus sadness).

6. Discussion

The principle finding is the asymmetry between the confirmations and the challenges. Secularization and the movement toward concreteness are exactly the claims one would expect a token-based measure to recover, and it recovers them cleanly. The Romantic nature turn and the industrial imagery move in the direction predicted by literary history but cannot clear a conservative significance bar—in the nature case because the eighteenth-century data are sparse, in the industrial case because the absolute rates are minute. The smallness is itself worth stating plainly: the industrialization of poetry is real as a direction but minute as a share of poetic attention within this canon.

The two challenged claims constitute the substantive contribution, with the coverage caveat firmly attached. If modern poetry is darker, sadder, or more intense, this corpus and these measures do not show it; arousal and explicit sadness both trend in the wrong direction. Three readings are compatible with the data. First, the claim may be incorrect for the canonical, public-domain slice of modern poetry, which terminates before the most disillusioned mid-century work and over-represents anthology pieces. Second, the EmoLex measure's 7.1% coverage may simply be too thin to detect a real trend. Third, modern mood may reside in syntax, irony, and imagery rather than explicit vocabulary of sad and high-arousal words—precisely what a bag-of-words rate cannot observe. Adjudicating among these is future work; the contribution here is to demonstrate that the explicit-vocabulary version of the claim fails, and to report it with an interval.

These reversals are consistent with prior quantitative findings rather than anomalous. Acerbi et al. (2013) report a general decline in the use of emotion-related vocabulary across twentieth-century books, and Morin and Acerbi (2017) document a two-century decline in emotional expression in Anglophone literature. The latter further observes that the decline is driven almost entirely by positive-emotion words, with negative-emotion words little changed—a pattern that explains why, in

the present corpus, the valence measure drifts downward while the explicit-sadness measure does not rise.

Methodologically, the study is a modest argument for two practices: bootstrapping whole poets rather than poems, which widens intervals honestly in sparse or poet-dominated eras (and which demotes the nature turn to weakly supported); and validating inexpensive instruments before trusting them, which is what permits the study to trust concreteness, hedge on valence, and flag the emotion measures.

7. Limitations

Word lists are handbuilt; the validation step exists precisely because they are, and the valence instrument in particular is only moderately reliable (era-level $r = 0.41$). Dating is by author flourish rather than composition date, so era boundaries are soft and a poet's output is pinned to a single year. The corpus is canonical and public-domain, terminating in the early twentieth century, so nothing here speaks to non-canonical, contemporary, or non-anglophone poetry. The emotion lexicon's coverage (7.1%) makes the joy, sadness, and fear measures the weakest in the study. Early eras are sparse, producing wide intervals that should not be over-interpreted. Finally, all measures are descriptive: the study counts and averages, and makes no claim about why any trend occurs.

8. Conclusion

Operationalising eight received claims about English poetry across approximately 34,000 poems yield a mixed and useful scoreboard: secularization and the turn to concreteness are confirmed; the Romantic nature turn, industrial imagery, the darker mood, and the visual turn are directionally correct but not decisively supported; and the claims that modern poetry is more emotionally intense are more explicitly sad are challenged by the data, subject to a real lexicon-coverage caveat. Each verdict is accompanied by a confidence interval and a public, reproducible pipeline, so any may be re-checked, contested, or improved with a better lexicon or a broader corpus. The challenged claims, not the confirmed ones, are the principle motivation for this kind of analysis.

Data and code availability

All code is openly available at [GitHub](#). The corpus is built entirely from public-domain sources: PoeTree English subset (Zenodo record 10907309) and PoetryDB ([PoetryDB.org](#)). The validation lexicons are third-party resources used under their respective licenses and are not redistributed; the repository documents where each may be obtained. The analysis is fully reproducible from the source data via the scripts described in Section 3.

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